

ROSALIND AGCM, AB, Canada

WMO: 712020

Lat: 52.7923N

Lon: 112.4228W

Elev: 2323

StdP: 13.50

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.7	-18.3	-29.7	1.1	-23.4	-24.3	1.6	-18.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	83.6	62.1	80.5	61.6	77.5	60.4	66.3	77.3	64.3	76.1	62.6	73.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	91.0	72.1	59.9	83.9	69.5	58.1	78.8	67.4	32.3	77.0	30.7	76.3	29.4	74.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.7	89.6	5.5	4.3	-33.7	92.6	-36.9	95.1	-40.0	97.5	-44.0	100.6		
(5)			-30.0	71.1	5.4	2.8	-33.9	73.1	-37.0	74.8	-40.1	76.4	-44.0	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	13.9	11.8	23.5	37.3	51.2	58.3	62.3	60.5	52.1	39.2	23.7	12.1		
	DBStd	21.55	14.68	15.09	13.52	9.52	7.92	5.27	4.66	5.46	8.19	8.83	12.87	13.91		
	HDD50	5874	1119	1069	820	390	83	6	0	2	72	349	790	1174		
	HDD65	10161	1584	1489	1285	831	429	209	111	156	389	800	1239	1639		
	CDD50	1243	0	0	0	9	121	254	380	329	136	14	0	0		
	CDD65	56	0	0	0	0	2	8	26	17	3	0	0	0		
	CDH74	1516	0	0	0	6	147	180	473	474	233	3	0	0		
	CDH80	316	0	0	0	0	23	22	100	111	60	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.7	0.8	0.5	0.7	1.2	1.9	3.1	3.0	2.2	1.3	0.7	0.6	0.6		
(15) Precipitation	PrecMax	20.8	2.5	1.0	1.5	2.4	3.3	4.5	6.5	4.2	3.0	1.8	2.2	1.2		
	PrecMin	11.4	0.0	0.1	0.1	0.2	0.6	1.0	0.9	0.2	0.1	0.2	0.1	0.2		
	PrecStd	2.8	0.6	0.3	0.4	0.6	0.8	1.0	1.3	0.9	0.8	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	43.1	57.2	72.4	83.1	82.4	87.1	86.7	86.8	73.0	58.5	43.3		
		MCWB	36.9	37.9	45.2	50.3	56.4	61.7	67.4	62.2	60.2	54.1	45.6	37.2		
	2%	DB	40.2	39.0	49.7	65.7	78.1	78.9	82.7	83.1	80.7	64.8	49.5	39.4		
		MCWB	35.3	34.3	40.4	47.7	54.7	60.6	65.2	62.5	58.1	50.5	40.8	34.5		
	5%	DB	37.6	36.1	43.6	59.4	73.8	75.4	79.4	79.3	75.4	59.4	43.7	35.6		
		MCWB	33.9	32.4	37.5	44.2	53.3	59.2	64.3	62.2	56.9	47.1	37.2	31.6		
	10%	DB	34.5	32.9	39.4	54.0	68.8	71.7	75.7	75.7	69.8	54.4	39.5	31.1		
		MCWB	31.2	30.2	34.7	42.0	51.2	57.4	62.5	61.0	54.4	44.6	34.9	27.9		
	0.4%	WB	37.6	37.9	45.7	51.8	58.4	65.7	70.7	68.3	62.0	55.3	46.9	37.8		
		MCDB	42.5	42.9	54.8	71.7	76.8	77.2	80.7	77.3	79.2	70.0	57.1	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	34.8	40.8	48.0	55.9	62.9	67.2	65.4	59.5	51.6	41.6	34.8		
		MCDB	39.7	38.4	48.6	64.2	73.3	74.4	78.0	76.9	76.7	62.1	48.2	39.1		
	5%	WB	34.1	32.7	37.5	45.2	54.3	60.6	65.3	63.4	57.3	48.2	37.9	31.7		
		MCDB	37.7	35.8	42.9	57.8	70.6	71.6	76.6	75.9	72.4	57.8	42.8	35.2		
	10%	WB	31.3	30.2	34.9	42.5	52.4	58.7	63.6	61.7	55.3	45.2	35.2	27.9		
		MCDB	34.5	32.9	39.0	52.7	66.9	69.1	74.4	73.7	68.4	53.5	38.6	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.8	18.5	21.8	25.0	22.2	23.6	25.2	25.8	21.9	17.5	16.1		
		MCDBR	18.2	19.7	23.0	33.0	33.2	28.1	29.7	33.1	37.1	30.9	22.8	18.3		
		MCWBR	14.7	15.9	15.3	17.2	14.5	12.7	15.4	15.8	17.1	17.4	15.1	14.6		
		MCDBR	17.8	18.8	22.0	30.3	29.2	25.3	27.4	28.8	34.0	28.5	20.7	18.1		
	5% WB	MCWBR	14.7	15.3	15.0	16.5	13.5	13.0	15.6	15.3	16.4	16.6	14.1	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.297	0.330	0.349	0.362	0.358	0.370	0.314	0.283	0.255	0.247		
	taud		2.280	2.358	2.392	2.363	2.365	2.352	2.396	2.340	2.466	2.530	2.454	2.302		
	Ebn at Noon		232	271	285	287	286	282	281	270	276	266	241	219		
	Edh at Noon		19	24	29	34	36	37	35	35	27	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		334	667	1127	1494	1825	1921	2000	1637	1135	671	335	245		
	RadStd		33	55	84	142	120	124	102	114	121	79	36	23		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (89 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page